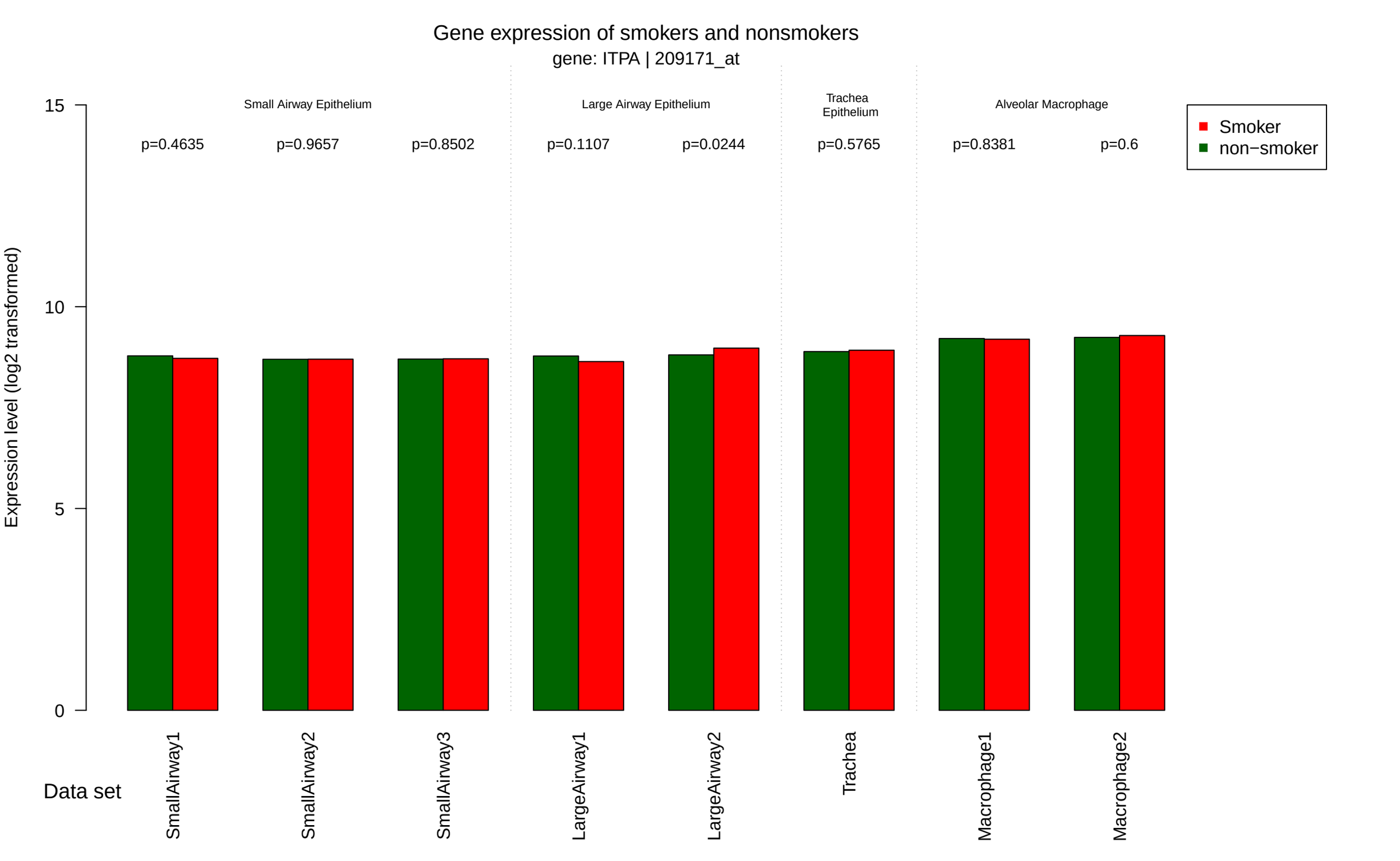
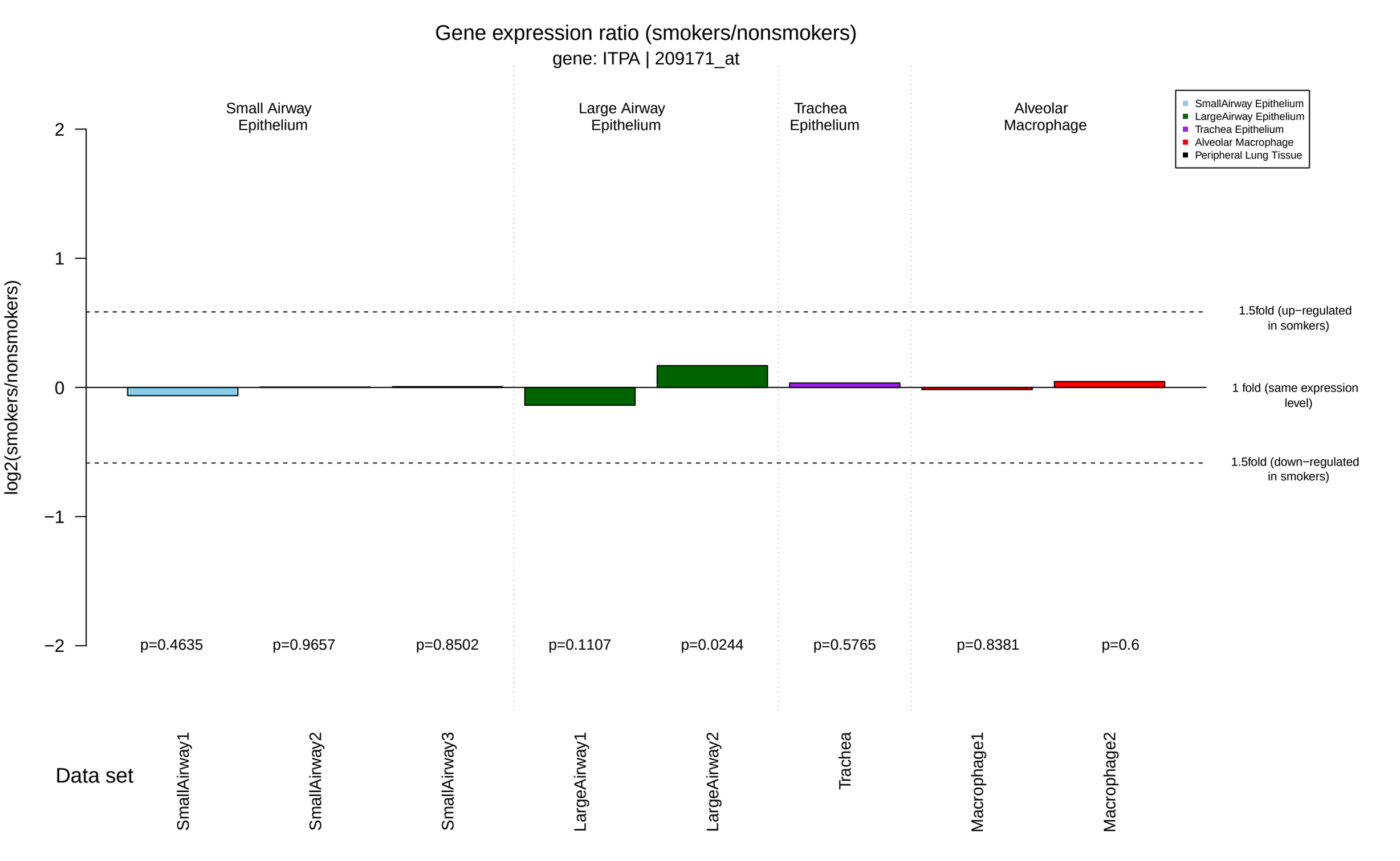
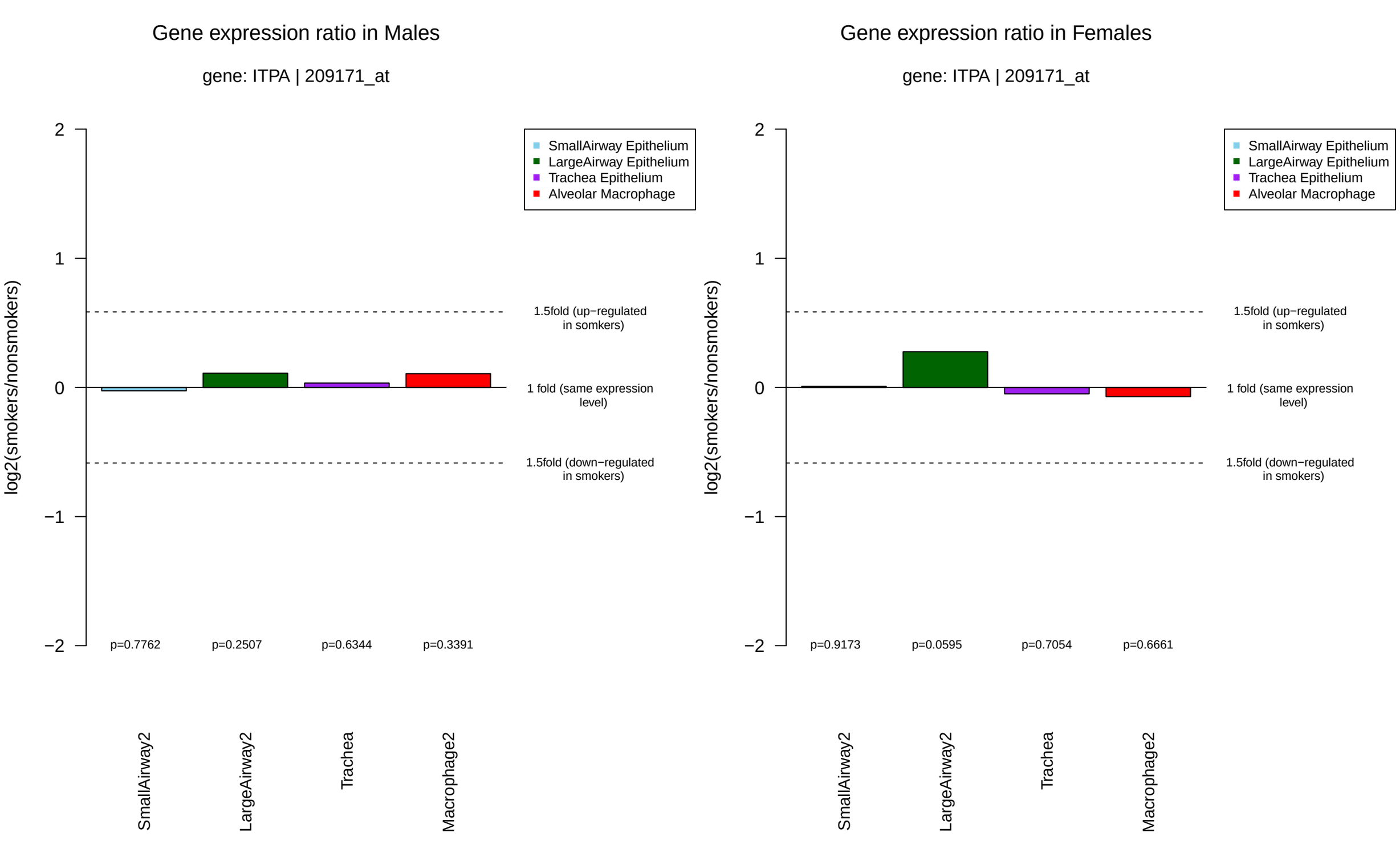
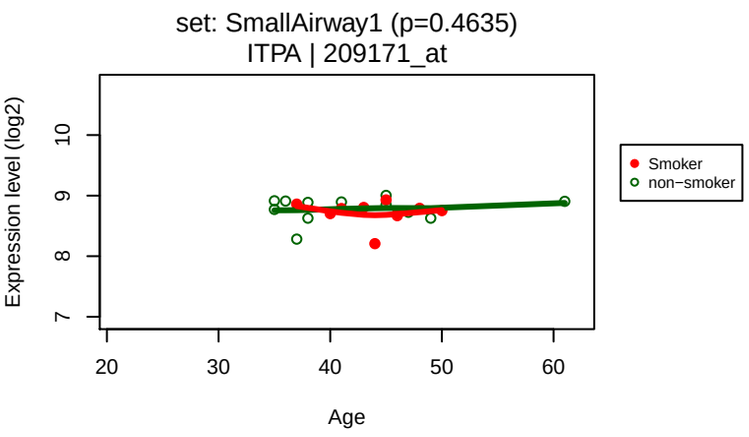
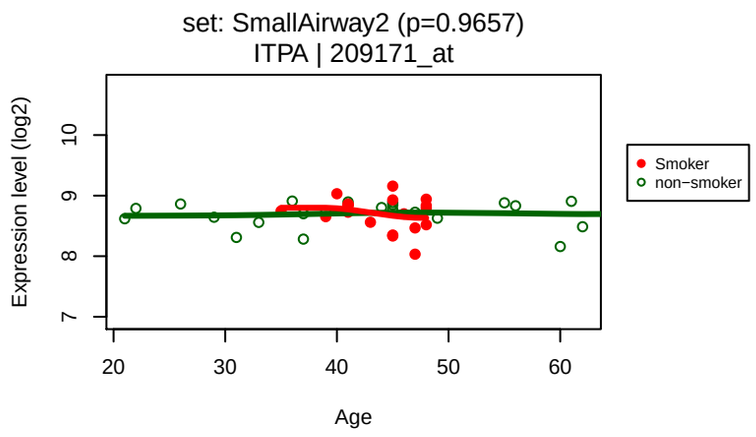
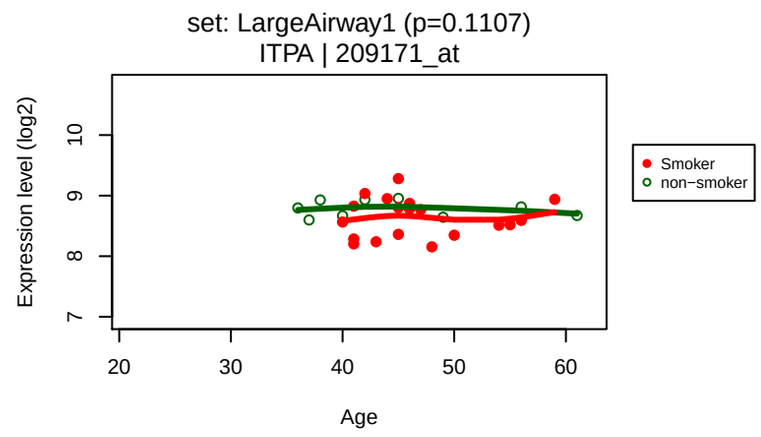

Gene expression of smokers and nonsmokers
gene: ITPA | 209171_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4635
p=0.9657
p=0.8502
p=0.1107
p=0.0244
p=0.5765
p=0.8381
p=0.6
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

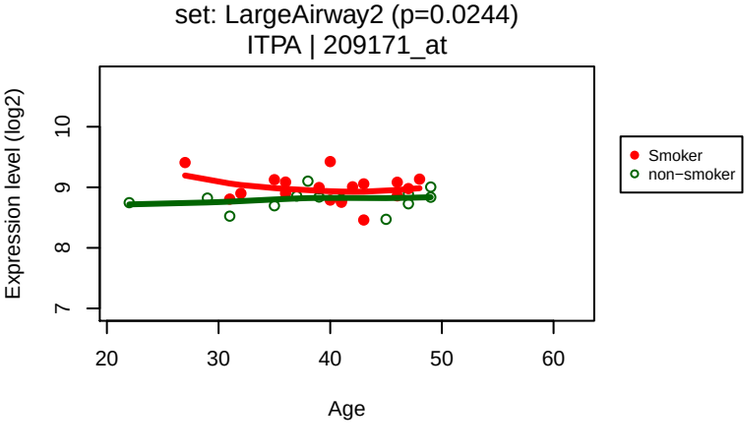

Gene expression ratio (smokers/nonsmokers)
gene: ITPA | 209171_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4635
p=0.9657
p=0.8502
p=0.1107
p=0.0244
p=0.5765
p=0.8381
p=0.6
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

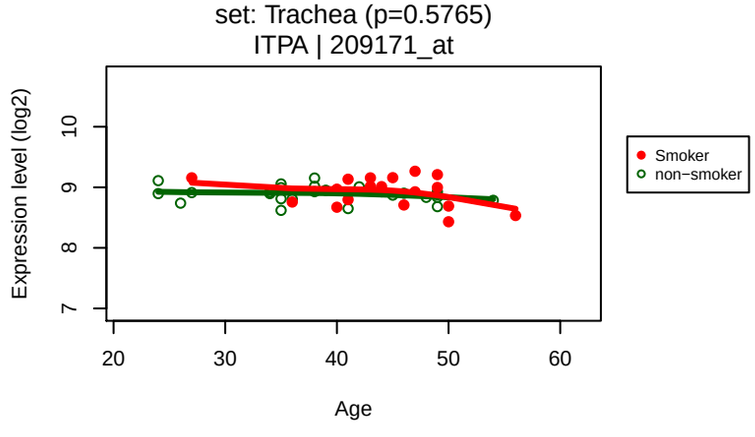

Gene expression ratio in Males
gene: ITPA | 209171_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7762
p=0.2507
p=0.6344
p=0.3391
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ITPA | 209171_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9173
p=0.0595
p=0.7054
p=0.6661
SmallAirway2
LargeAirway2
Trachea
Macrophage2

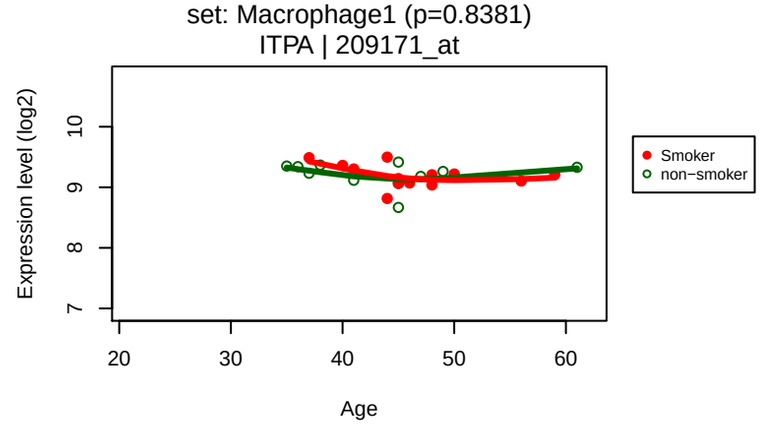

set: SmallAirway1 (p=0.4635)
ITPA | 209171_at
Expression level (log2)
Age
Smoker
non−smoker

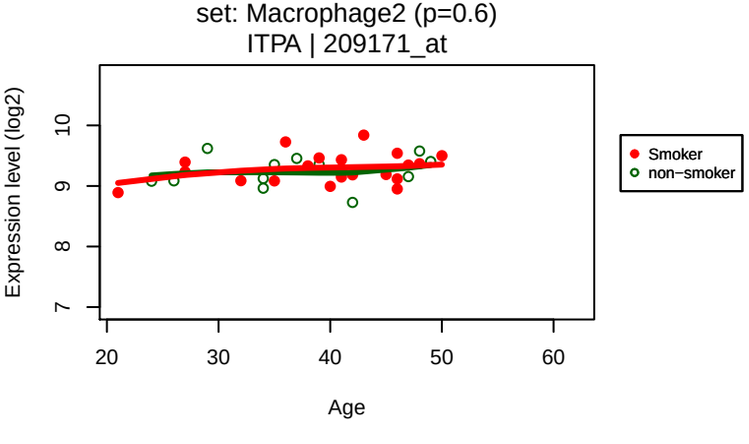

set: SmallAirway2 (p=0.9657)
ITPA | 209171_at
Expression level (log2)
Age
Smoker
non−smoker

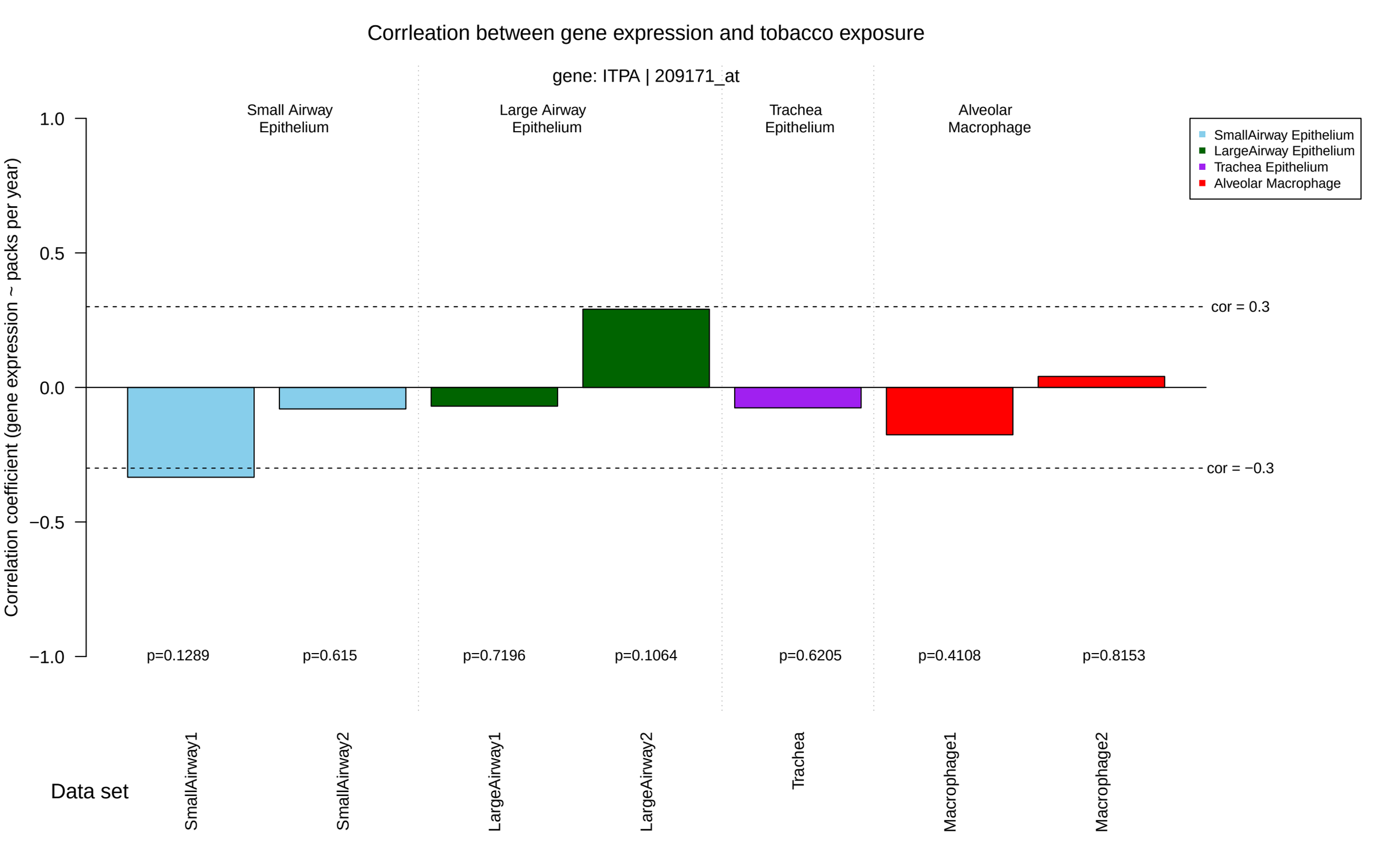

set: LargeAirway1 (p=0.1107)
ITPA | 209171_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0244)
ITPA | 209171_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.5765)
ITPA | 209171_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8381)
ITPA | 209171_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6)
ITPA | 209171_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ITPA | 209171_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.1289
p=0.615
p=0.7196
p=0.1064
p=0.6205
p=0.4108
p=0.8153
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2